Rates Market Maker

Inventory risk management in a simulated bond market-making desk

Methodology report: approach, design choices and justification of the inventory skew

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> **Educational project.** This is a learning exercise built to understand how a rates market maker thinks about spread, inventory and risk. It is not a trading system. Everything is simulated: the market is a random walk, the clients are a stylised probabilistic model, and the parameters were set by hand. Real desks use far richer models, real data and execution infrastructure that are out of scope here. The results show how the mechanisms behave inside this toy world; they say nothing about live-market profitability.

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1. Introduction and objectives

A market maker continuously quotes a price at which it will buy (the **bid**) and a price at which it will sell (the **ask**). The gap between the two, the **spread**, is its revenue for providing liquidity. In exchange it takes on a risk: every client trade changes its inventory, and the value of that inventory moves with the market. For a rates desk, the relevant risk is the sensitivity of the book to interest rates, measured by the **DV01**.

This project builds a small simulator to study one question: **how should a market maker adjust its quotes to keep its inventory risk under control, and what does that control cost in expected profit?**

The project compares three ways of managing risk, on identical simulated markets and identical client behaviour:

- **Strategy A**: a fixed spread around the fair price, with no risk management (the baseline).
- **Strategy A + hard DV01 limit**: the same quotes, but the desk refuses any trade that would push its absolute DV01 above a limit.
- **Strategy B**: an **inventory skew**, where the centre of the quotes is shifted against the current position so that the desk is more likely to trade back toward flat. It is tested alone and combined with the hard limit.

1.1 What this report is and is not

The objective of the project was learning: building each component, understanding why it behaves as it does, and being able to defend each choice. This report documents the reasoning behind the design, with particular attention to the choice of the skew parameter, which is the most delicate decision in the project.

It is important to be clear about the distance between this simulator and a professional desk. Real market makers work with real order flow, several instruments and curves, hedging in liquid futures and swaps, adverse-selection models, funding and capital costs, and optimal-control or machine-learning approaches to quoting. The simulator keeps none of that. What it keeps is the core trade-off (spread revenue against inventory risk) in a form small enough to understand completely and to test rigorously. Section 8 lists the simplifications explicitly.

2. Background: market making and inventory risk

2.1 Spread, fair price and the source of profit

Let the **fair price** be the best estimate of the bond's value at a given moment. A market maker quotes a bid below it and an ask above it. If a client buys at the ask and another sells at the bid, the desk earns the spread without taking a view on the market. A wider spread earns more per trade but attracts fewer trades, because clients prefer better prices. This is the first trade-off in the model.

2.2 Inventory risk

Trades do not arrive in matched pairs. After a client sells to the desk, the desk is long the bond and loses money if rates rise (bond prices fall). The spread earned on each trade is small compared with the price moves that can happen while the position is held. Left alone, inventory performs a random walk, and the P&L of a pure spread-capture strategy becomes noisy and occasionally negative. This is the second trade-off: more trading means more spread income, but also more inventory to manage.

2.3 DV01

DV01 (dollar value of one basis point) is the change in value of a position for a one-basis-point change in yield. It converts positions in different instruments into a common measure of rate risk, which is why desks set limits in DV01 rather than in notional. In this project it is computed by bumping the yield by 1 bp and repricing; for the 10-year bond used here it is about 0.0811 per 100 of face value, i.e. roughly 811 per 1 million of position.

3. The model

Every component is deliberately simple, so that each one can be explained and tested in isolation.

3.1 Interest rates

The yield follows an arithmetic random walk with constant volatility:

$$r(t+1) = r(t) + \sigma \cdot \varepsilon(t), \quad \varepsilon(t) \sim N(0,1), \quad \sigma = 0.15 \text{ bp per step}$$

Each path has 50 steps, so the standard deviation of the total rate move is about 1.05 bp. The step has no precise calendar meaning; it is a unit in which quotes are refreshed and at most one client arrives on each side. The model is stylised on purpose: it has no mean reversion, no jumps and no volatility clustering.

3.2 Bond price and DV01

The instrument is a 10-year bond with a 4% coupon, priced as the sum of discounted cash flows at the simulated yield, starting from an initial yield of 4% (so the bond starts near par). The price is quoted per 100 of face value. DV01 per 100 is obtained by bumping the yield by 1 bp:

$$DV01 \text{ per } 100 = P(r - 1bp) - P(r)$$

The DV01 of a position of size N (face value) is then:

$$DV01(\text{position}) = (N / 100) \cdot DV01 \text{ per } 100$$

DV01 per 100 itself changes slightly as rates move (convexity), which matters for the limit check described in section 5.2.

3.3 Quotes

With a fair price F and a half-spread h = 0.02, the baseline quotes are:

$$bid = F - h, \quad ask = F + h$$

3.4 Client behaviour

Clients are modelled probabilistically. At each step, a client may want to buy from the desk (at the ask) and another may want to sell to it (at the bid). The probability that a client trades depends on how unattractive the quote is compared with the fair price:

$$p = \min(1, \text{flow} \cdot \exp(-\text{sensitivity} \cdot \text{gap}))$$

where gap_buy = ask − F and gap_sell = F − bid, flow = 0.3 and sensitivity = 30. If the quote equals the fair price, the trade probability is 0.30. With the baseline half-spread of 0.02, it falls to about 0.165 per side per step, which gives roughly 16.5 trades per path on average. A quote better than fair is capped at probability 1.

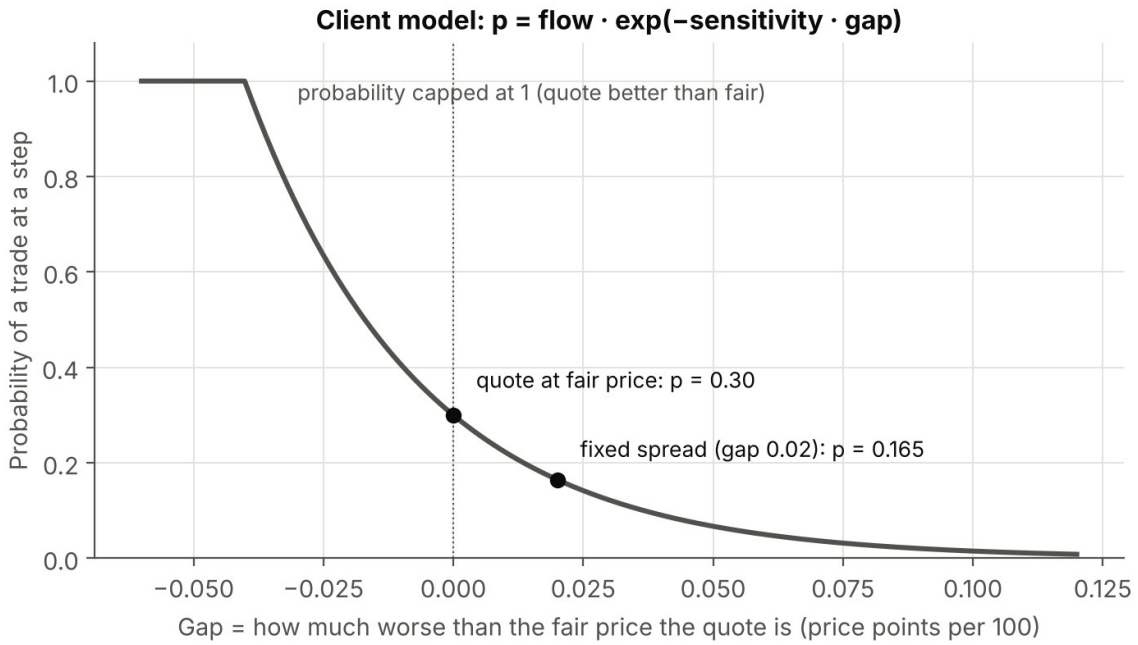


Figure 1. Client model. A quote that is worse than the fair price loses clients exponentially. This is what makes a wider spread both more profitable per trade and less frequently hit.

This exponential form is the simplest function that captures the economic intuition (better price, more flow). Its two parameters were set by hand, not estimated from data, and the conclusions depend on them: a higher price sensitivity penalises skewing more, a lower one makes it cheaper. Section 8 returns to this.

3.5 Position, cash and P&L

Trade size is fixed at 1 million of face value, with at most one client buy and one client sell per step. When a client buys, the desk sells and its position falls; when a client sells, the desk buys and its position rises. Cash is updated at the traded price, and the P&L is marked to market at the fair price:

$$P\&L(t) = cash(t) + (position(t) \,/\, 100) \cdot F(t)$$

Because the desk can be long or short, position, cash and DV01 are all tracked for every step of every path.

4. Implementation and validation

4.1 Vectorised simulation

The simulator is written in Python with NumPy. All paths are simulated at once, and the time loop only iterates over the 50 steps, never over the paths. Pricing the bond for all paths and steps with vectorised cash-flow discounting is about 20 times faster than a straightforward loop implementation (measured at 20.7x), which makes it practical to run the thousands of paths needed to estimate tail statistics.

4.2 Common random numbers

All strategies are run on **exactly the same** simulated rate paths and the same uniform random draws for client arrivals (u_buy, u_sell). A client trades when its uniform draw is below the trade probability

computed from the strategy's quote. Strategies therefore differ only through their quoting and risk rules. This technique, known as common random numbers, removes most of the sampling noise from comparisons: a difference between two strategies is caused by the strategies, not by luck in the draws. All seeds are fixed, so every result is reproducible.

4.3 Testing

Each function was tested before being used in the next step. The checks cover, for example, agreement between the vectorised and loop bond pricers, the sign and magnitude of DV01, bid below ask, correct cash and position updates on a buy and on a sell, the direction of the skew shift for long and short positions, and the behaviour of the hard limit (trades that increase risk beyond the limit are refused, trades that reduce risk are always accepted). Appendix B lists the functions.

5. The strategies

5.1 Strategy A: fixed spread

Strategy A quotes F ± 0.02 at all times and accepts every client trade. It is the benchmark: it earns the spread but does nothing about inventory. Figure 2 shows 500 paths of strategy A. The average P&L grows roughly linearly, which is the spread income, but the fan of paths widens over time and some paths end negative. The position DV01 wanders freely in steps of about 811, and on some paths reaches several times that.

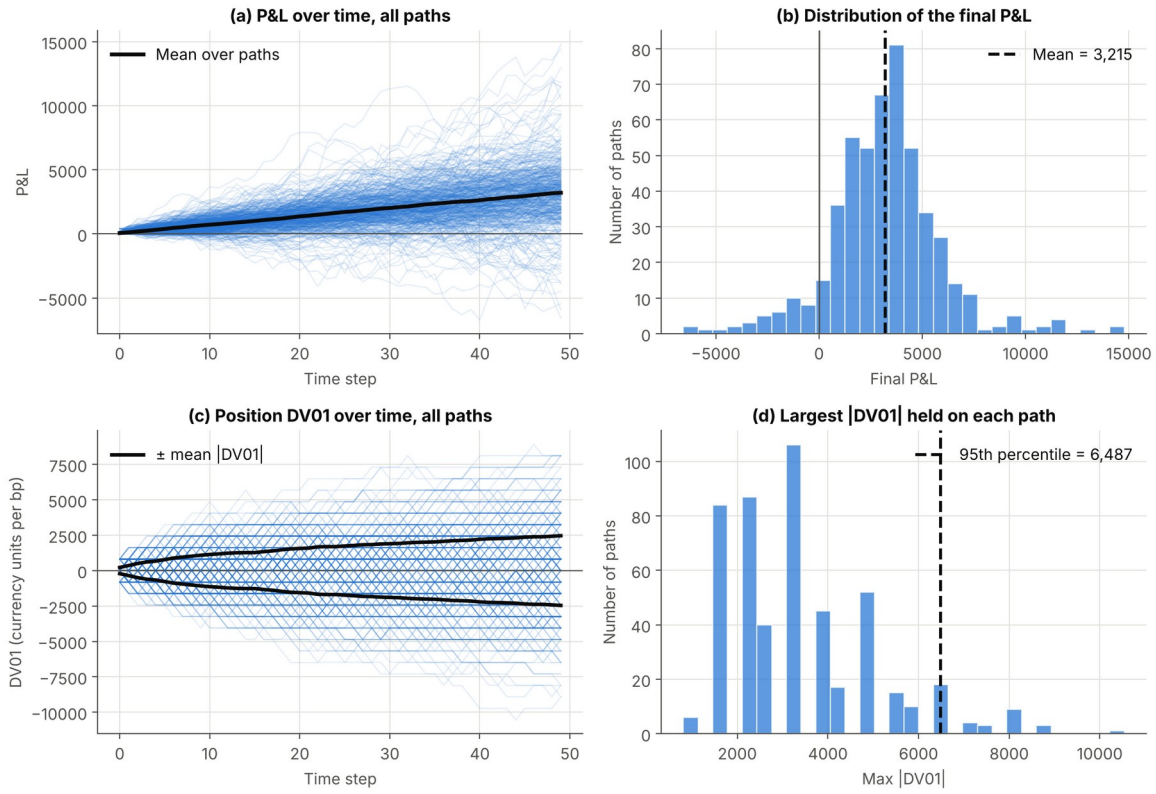


Figure 2. Strategy A on 500 paths (first simulation). (a) P&L paths and their mean. (b) Distribution of the final P&L. (c) Position DV01 over time. (d) Largest absolute DV01 reached on each path, with the 95th percentile.

On this baseline, roughly 80% of the paths reach a position of 3 million or more at some point, so the risk is not an occasional accident: it is the normal behaviour of the strategy.

5.2 Strategy A with a hard DV01 limit

The first way to control risk is a rule: **refuse any trade that would push the absolute DV01 above a limit** (3,200, about four 1-million trades). A trade that reduces the absolute DV01 is always accepted. The limit is checked at trade time, so the DV01 can drift slightly above the limit afterwards, since DV01 per 100 moves with rates.

This rule is easy to explain and easy to audit, and it is how many risk frameworks work in practice. Its weakness is that it is binary: the desk quotes exactly the same prices when it is flat as when it is one trade away from the limit, and it simply turns away the trades that would breach it. It reduces the tail but does nothing to attract the trades that would bring the position back.

5.3 Strategy B: inventory skew

The second approach changes the **prices** instead of refusing trades. If the desk is long, it would like to sell, so it lowers both quotes: the lower ask attracts clients who want to buy, and the lower bid discourages clients who want to sell to it. If it is short, it raises both. The shift is proportional to the position's DV01:

$$shift = skew \cdot DV01(position) \,/\, 1000$$

$$bid = F - h - shift, \qquad ask = F + h - shift$$

The skew parameter is expressed in price points per 1,000 of DV01. Three design decisions deserve explanation.

- **The shift is proportional to DV01, not to notional.** DV01 is the quantity the desk actually wants to control, and it keeps the rule meaningful if the instrument or the rate level changes.
- **Both quotes move by the same amount.** The spread width stays at 2h, so the skew changes only the centre of the quotes. This isolates the effect of inventory management from the effect of the spread level (a dynamic spread is a separate, optional extension; see section 9).
- **The relationship is linear.** It is the simplest rule that has the right sign, is symmetric for long and short positions, and has a single parameter that can be calibrated. It is a heuristic, not the solution of an optimal-control problem.

The skew has a cost. When the desk is long, its ask can fall below the fair price, meaning that it sells at a loss in expected terms in order to reduce risk. This is the price paid for risk reduction, and the calibration in section 6 is about deciding how much of it is worth paying.

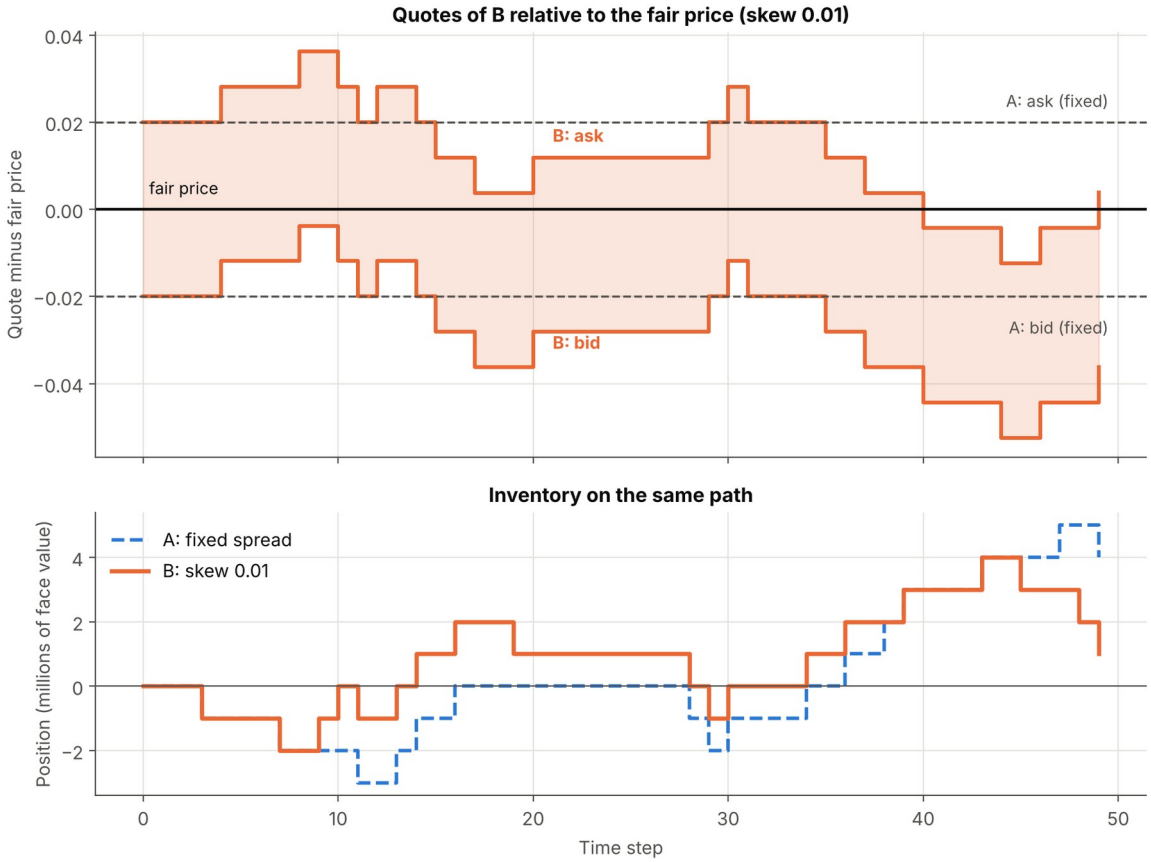


Figure 3. The skew mechanism on one path (skew 0.01). Top: B's quotes relative to the fair price, against A's fixed quotes (dashed). When the position grows, both quotes shift down. Bottom: the position of A and B on the same path; the two strategies diverge only once the skew starts to act, and B ends with a smaller inventory (+1M versus +4M).

On this path the final position is +4 million for A and +1 million for B, and the final P&L is −3,174 for A against −918 for B. One path is an illustration, not evidence; the statistics in sections 6 and 7 rest on thousands of paths.

6. Choosing the skew

The skew is the central parameter of Strategy B. A skew of zero reproduces A (with the limit), and a very large skew pushes the quotes so far from fair that the desk gives away its edge to stay flat. The right value lies somewhere in between, and the objective of this section is to find it with a justification that does not depend on taste.

6.1 The trade-off

Raising the skew reduces risk (lower P&L standard deviation, lower DV01) but also reduces expected profit, because the desk trades at worse prices when it is skewed. There is no skew that is best on every measure, so a choice requires a criterion. To avoid choosing the criterion after looking at the results, it was fixed in advance:

> **Criterion.** Among the skew values tested, choose the one that reduces risk the most while giving up at most 5% of the mean P&L of the benchmark (strategy A with the hard limit).

The benchmark is A with the limit rather than A alone because that is the realistic alternative a desk would use. The 5% budget is a judgement about risk appetite, and it is the one subjective element of the method; section 6.4 shows how sensitive the result is to it.

6.2 The calibration grid

Eleven skew values from 0 to 0.04 were run with the hard limit at 3,200, on 5 independent simulations of 500 paths, and the results were averaged. Table 1 and Figure 4 show the outcome.

Table 1. Skew grid with the hard limit (average of 5 simulations; 500 paths × 50 steps each).

Skew	Mean P&L	Std P&L	5th pct P&L	Mean | DV01|	Δ mean vs 0	Δ std vs 0
0 (A + limit)	2,978	1,686	100	1,147	–	–
0.0025	3,022	1,556	417	1,023	+1.5%	−7.7%
0.005	3,006	1,429	618	913	+0.9%	−15.3%
0.0075	2,956	1,309	899	816	−0.7%	−22.4%
0.01	2,875	1,215	974	733	−3.5%	−28.0%
0.0125	2,787	1,147	1,017	665	−6.4%	−32.0%
0.015	2,690	1,081	993	610	−9.7%	−35.9%
0.02	2,445	984	898	527	−17.9%	−41.7%
0.025	2,178	883	761	462	−26.9%	−47.6%
0.03	1,902	801	671	413	−36.1%	−52.5%
0.04	1,286	733	135	341	−56.8%	−56.5%

Highlighted row: the chosen skew. Std and 5th percentile of the final P&L are in currency units; Mean |DV01| is the average absolute position DV01 over time and paths.

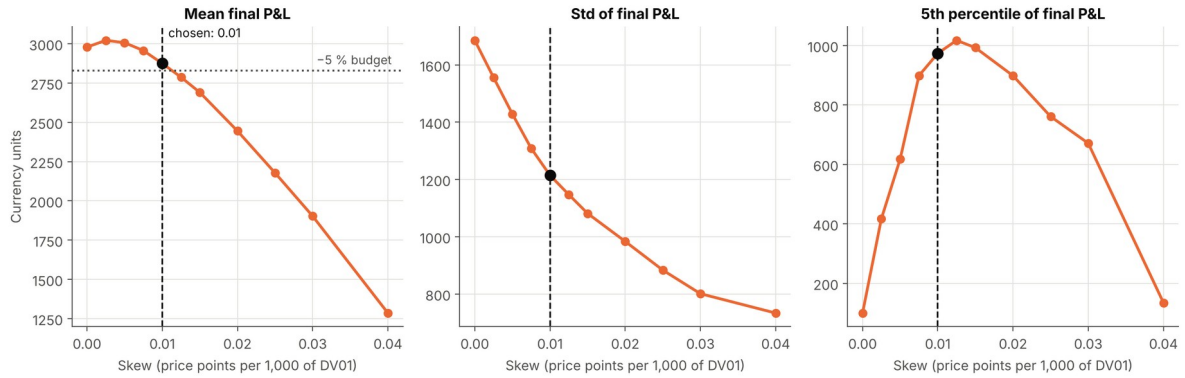


Figure 4. Mean, standard deviation and 5th percentile of the final P&L as a function of the skew (hard limit 3,200, average of 5 simulations). Dashed line: chosen skew 0.01. Dotted line: the −5% budget on the mean.

6.3 Reading the grid

- **Small skews are almost free.** Up to about 0.005 the mean P&L does not fall at all (it even rises slightly, which is within noise), while the standard deviation falls by 15%. Moderate inventory control is cheap because the position spends time near zero, where the shift is small.
- **The cost accelerates.** The mean P&L falls slowly at first and then faster: the larger the skew, the more often the desk trades at quotes that are worse than fair. Going from 0.01 to 0.02 costs 14 more points of mean P&L for 14 more points of risk reduction; going from 0.02 to 0.04 costs 40 more points for 15.

- **Risk reduction has diminishing returns.** The standard deviation falls more and more slowly because the remaining variance comes from the rate moves on the small positions that are still held.
- **The 5th percentile peaks.** It rises to about 1,017 at a skew of 0.0125 and then falls, reaching only 135 at 0.04. Beyond the peak, the desk is giving away so much edge that the bad paths get worse again: too much skew is itself a source of risk.

Under the criterion, the largest skew within a 5% mean-P&L budget is **0.01** (−3.5% mean P&L, with −28% standard deviation). The next grid point, 0.0125, costs 6.4% and is excluded. The 5th-percentile optimum (0.0125) lies right next to the chosen value, which is reassuring: the criterion and a pure tail-risk view point to the same neighbourhood.

6.4 Is the choice an artefact of the sample?

Choosing a parameter on the same data used to evaluate it risks overfitting. Two checks were run.

- **Out-of-sample check.** The 5 simulations were split into a calibration group (simulations 1 to 3) and a test group (simulations 4 and 5). The criterion selects 0.01 in both groups: at 0.01 the mean P&L cost is −2.9% (calibration) and −4.4% (test), while at 0.0125 it is −5.9% and −7.3%, above the budget in both cases. The choice does not depend on which simulations are used.
- **Sensitivity to the budget.** The 5% budget is a risk-appetite judgement. A tighter budget (about 1%) would select a skew near 0.0075; a looser one (about 10%) would select 0.015. Over this range the standard deviation reduction varies between 22% and 36%. The qualitative conclusion, that skewing is effective, does not depend on the exact value; the quantitative recommendation does depend on the desk's risk appetite, as it should.

The choice of 0.01 should therefore be read as: *given a desk willing to give up about 5% of expected profit for lower risk, this is the skew that achieves the largest risk reduction in this model*. It is a statement conditional on the model, the parameters and the criterion.

7. Results and robustness

7.1 Final comparison

Table 2 compares the four configurations (A, A with the limit, B at skew 0.01, B at skew 0.01 with the limit), averaged over 5 independent simulations of 500 paths each, using the same random numbers across strategies within a simulation.

Table 2. Final comparison (average of 5 simulations).

Metric	A	A + limit	B (skew 0.01)	B + limit
Mean final P&L	3,236	2,978	2,921	2,875
Std of final P&L	2,459	1,686	1,261	1,215
5th percentile of P&L	−628	+100	+984	+974
Share of losing paths (%)	7.1	5.0	1.2	1.2
Mean of max |DV01|	3,374	2,257	1,969	1,897
95th pct of max |DV01|	6,485	2,436	3,242	2,435
Mean |DV01|	1,587	1,147	748	733

Std and 5th percentile in currency units. Higher is better for mean and 5th percentile; lower is better for the other rows.

The main findings are:

- **The hard limit alone reduces the tail strongly.** It cuts the standard deviation by 31% and the 95th percentile of the maximum DV01 from 6,485 to 2,436, for an 8% lower mean P&L. It is effective at capping the worst case, but it leaves the typical position large (mean |DV01| 1,147).
- **The skew reduces typical risk much more than the limit.** At the same hard limit, adding the skew cuts the standard deviation by a further 28% and the mean |DV01| by 36%, for 3.5% lower mean P&L. The 5th percentile rises from +100 to +974, and the share of losing paths falls from 5.0% to 1.2%.
- **The skew alone is already close to the combination.** Without the limit, the skew cuts the standard deviation by about 25% compared with the hard limit alone for about 2% less P&L, though its 95th percentile of maximum DV01 (3,242) is higher, because without the limit nothing prevents a rare run of one-sided trades.
- **The two tools are complementary, not substitutes.** The skew manages the typical position, the limit protects against the unusual one. Their combination has the lowest risk on almost every measure.

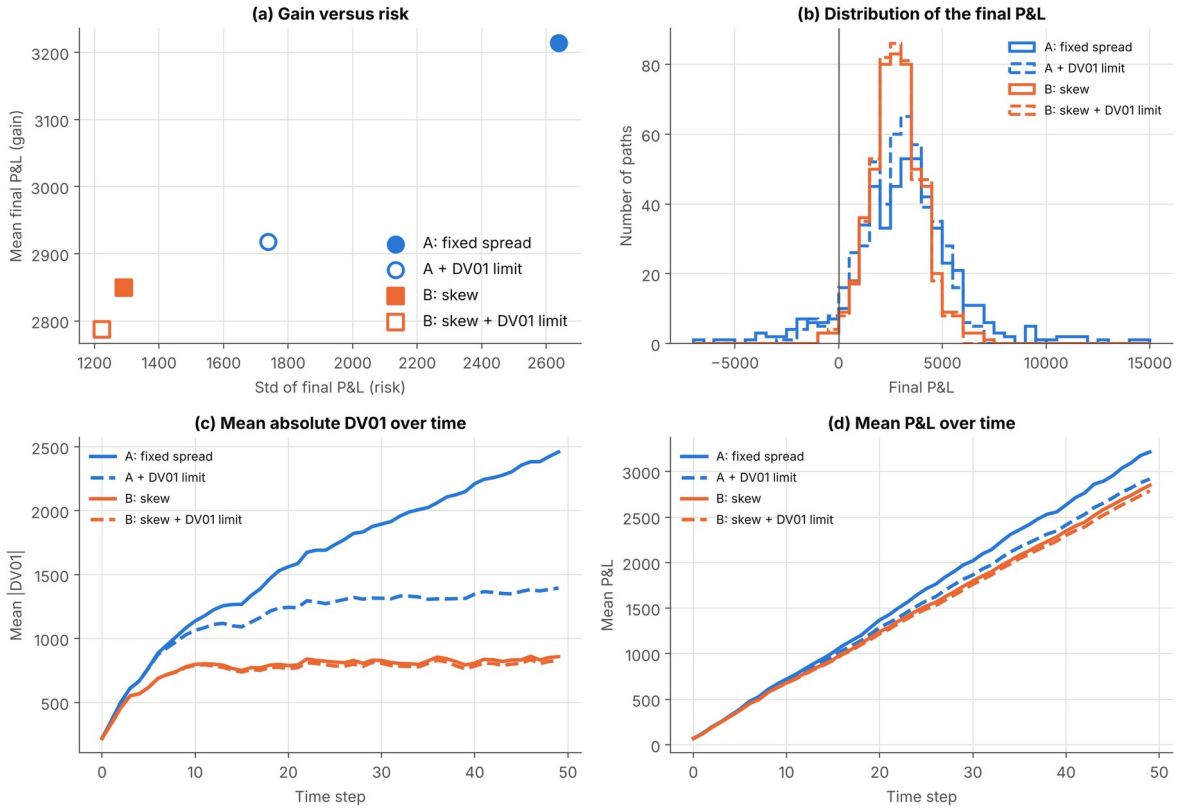


Figure 5. Final comparison on the first simulation (seeds 1 and 2), so values differ slightly from the 5-simulation averages in Table 2. (a) Gain versus risk: filled markers without limit, hollow with limit. (b) Distribution of the final P&L. (c) Mean absolute DV01 over time. (d) Mean P&L over time. Blue: A family; orange: B family; dashed: with hard limit.

Figure 5(a) is the most informative summary. Moving from A toward B shifts the point down (less P&L) and strongly to the left (less risk). The slope of that move is the efficiency of the risk-reduction tool: here, B gives up little P&L for a large reduction in risk, while the limit alone gives up more P&L for a smaller reduction. Panel (c) shows the mechanism at work: under A, the average absolute DV01 keeps growing with time, while under B it stabilises, because the skew pulls the position back.

7.2 How much of this is noise?

With 500 paths per simulation, a single simulation can mislead, especially for tail statistics such as the 5th percentile or the share of losing paths. To quantify this, the table above is the average of five independent simulations, and the standard deviation of each statistic across those five simulations was also computed.

- **Differences in mean P&L between strategies are small compared with their noise.** The mean P&L varies by about 70 to 110 between simulations. The difference between A and A + limit (about 260) is larger than that noise; the differences between A + limit and B (about 60) and between B and B + limit (about 50) are not. The statement that the skew costs about 3.5% of mean P&L should therefore be read as 'a small cost of the order of the noise', not as a precise figure.
- **Differences in risk are much larger than the noise.** For example, the standard deviation of the final P&L varies across simulations by 127 for A, 46 for A + limit, and only 5 for B with the limit, against differences between strategies of several hundred. The risk-reduction conclusions are robust.
- **Tails are the least precise.** Percentiles and loss frequencies need many paths. The ranking of strategies is stable, but the second significant digit of such statistics should not be trusted.

8. Limitations and what real desks do differently

This section is as important as the results. The simulator is a controlled experiment, and its simplifications determine what can and cannot be concluded.

Simplification in this project	What a real desk faces or does
No adverse selection: clients trade at random, independent of future price moves.	Informed clients trade when the quote is stale or wrong. Spread and skew must also protect against being picked off; this usually dominates the design of real quoting.
The fair price is known exactly.	The fair price is itself an estimate built from other quotes, curves, futures and flows, and is noisy.
One bond, one yield, one risk factor.	A book of many instruments across the curve, with curve, basis and convexity risks, and cross-hedging.
No hedging.	Inventory is hedged in liquid futures, swaps or on-the-run bonds. Skewing and hedging are used together.
Rates are a random walk with constant volatility.	Volatility changes, jumps around announcements, mean reversion and correlation between instruments.
Hand-set parameters; clients react to price through a simple exponential.	Client behaviour is estimated from data and differs by client type, size and market conditions. The results depend on the price sensitivity.
Fixed trade size, one buy and one sell at most per step, undefined step length.	Variable sizes, quote sizes depending on risk, continuous time and latency.
No funding, transaction, capital or balance-sheet costs.	These costs shape which inventory is worth holding.
Linear skew heuristic.	Optimal quoting is usually derived from stochastic control (for example, the Avellaneda–Stoikov framework) or learned, and quotes depend on time, volatility and the full risk profile.
500 paths per simulation, 5 simulations.	Tail estimates require many more paths; the mean P&L differences between B variants are within noise.
Limit checked at trade time only.	Limits are monitored continuously, intraday and overnight, with escalation procedures.

Two of these deserve further comment. First, the **absence of adverse selection** is the most favourable assumption for the market maker: in this model the spread is pure income. In real markets, part of the spread compensates for losses to informed flow, and an inventory skew can interact with it in ways this model cannot show. Second, the **parameter dependence** is not just a caveat but a testable property: the conclusions about the cost of skewing depend on the client's price sensitivity, and a proper sensitivity analysis (varying the volatility, flow and sensitivity) would show how robust the recommended skew is. It was identified as a next step and not completed.

9. Extensions and conclusion

9.1 Possible extensions

- **Dynamic spread (Strategy B2).** Widening the half-spread with the absolute DV01, h(t) = h + slope · |DV01| / 1000. A first quick test (skew 0.01 and slope 0.0025, limit 3,200) gave mean P&L 3,030, std 1,273, 5th percentile 1,042 and mean |DV01| 762, against 2,875, 1,215, 974 and 733 for B with the limit. In this test the dynamic spread acts mostly as a way to recover profit and gives a modest improvement of the gain-versus-risk frontier (about +2.5% gain at equal risk). It needs its own two-dimensional calibration with a pre-specified criterion, so this is preliminary evidence only.
- **Sensitivity analysis** on volatility, client flow and price sensitivity, to measure how robust the chosen skew is.
- **More realistic rates**: a Vasicek model, real data from public sources (for example FRED), and shocks to the curve.
- **Hedging** of the inventory with a second instrument, and a model of **adverse selection**.
- **Optimal quoting**: comparing the linear skew with a solution from stochastic control such as Avellaneda–Stoikov.

9.2 Conclusion

In this simulated environment, controlling inventory through quotes is more efficient than controlling it by refusing trades. At the same hard limit, a skew of 0.01 reduces the standard deviation of the final P&L by 28% and the mean absolute DV01 by 36%, for about 3.5% lower mean P&L; the share of losing paths falls from 5.0% to 1.2%. The skew was chosen by a criterion fixed before the results were seen, validated out of sample, and its sensitivity to the risk budget was analysed.

The wider lesson of the project is not the particular numbers but the method: build the model in small tested pieces, compare strategies on common random numbers, pre-specify the selection criterion, separate calibration from evaluation, quantify the noise, and state the limits of the model openly. The results hold in this toy world, and the interest of the exercise is in understanding why.

Appendix A. Parameters

Parameter	Value	Meaning
n_paths / n_steps	500 / 50	Paths per simulation and steps per path
Simulations	5	Independent repetitions for robustness
sigma	0.15 bp per step	Volatility of the yield random walk
Bond	10 years, 4% coupon	Instrument; DV01 per 100 ≈ 0.0811
half_spread	0.02	Half-spread in price points per 100
flow	0.3	Trade probability when the quote equals fair
sensitivity	30	Decay of the trade probability with the gap
trade_size	1,000,000	Face value per client trade
dv01_limit	3,200	Hard limit on absolute position DV01
skew (chosen)	0.01	Price points per 1,000 of DV01
Skew grid	0 to 0.04	11 values; see Table 1

Appendix B. Code structure

Function	Role
bond_price, bond_price_loop	Vectorised and loop bond pricers (the loop version is the reference for testing)
calculation_DV01	DV01 per 100 by a 1 bp bump
simulate_rates	Arithmetic random-walk yield paths for all paths at once
quote_fixed_spread	Fixed-spread bid and ask
quote_skewed	Bid and ask shifted by the inventory skew
client_probabilities	Trade probabilities from the gap to the fair price
draw_client_uniforms	Uniform draws shared across strategies (common random numbers)
run_strategy_a	Baseline simulation, optional hard DV01 limit
run_strategy	General simulation with skew and optional hard limit; returns position, cash, P&L, bid and ask
position_dv01	DV01 of a position at each step
summarise, describe_summary	Statistics: mean, std, 5th percentile, share of losing paths, DV01 measures

The full code, the README and the figures are in the project repository.